PHY-112 | Principles of Physics-2

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Spring 2025 | Class #8

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Recap of the previous class!

What we studied in Class #7

Refer to Class #7 slides for details!

- What work done means for a charged particle in electric fields
- What are the natural direction of displacement for charged particles
- What does potential energy have to do with external charges
- How does a change in potential energy cause a change in kinetic energy for the moving particle
- How do we realize electric forces from electric potential

What do we need from Class 7!

Recall Class-7 (To understand Class-8)
General Definitions of Work done and Electric Potential Energy

- Work done due to moving a charge q_0 in an $\vec{E}$-field

$$W_{i\to f} = q_0 \int_{\text{start}}^{\text{final}} \vec{E}\cdot d\vec{l} = q_0 \int_{\text{start}}^{\text{final}} E dr$$

- Electric Potential Energy exchanged to moving a charge q_0 in an $\vec{E}$-field

$$\Delta U_{\text{elec}} = -q_0 \int_{\text{start}}^{\text{final}} \vec{E}\cdot d\vec{l} = -q_0 \int_{\text{start}}^{\text{final}} E dr$$

Electric Potential and where to find them!

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Electric Potential

Another way to look at an electric field source

Electric Potential —

- is a fixed property of the electric charge; Observer independent!
- exists everywhere in space, just like $\vec{E}$-fields.
- causes external/incoming/observer charges to have potential energy.
- is defined as

$$\Delta V = \frac{\Delta U}{q_0} = \frac{U_f - U_i}{q_0} = \frac{-(W_{i\to f})}{q_0}$$

$$\equiv \text{Potential Difference} \Longleftrightarrow \text{Voltage}$$

- is a scalar, no dealings with directions.
- is measured in J C^{-1}, or more commonly, Volt [V].

Electric Potential
The Hill Convention

Naturally speaking,

$$\text{Positive Charge} \longrightarrow \text{Potential Downhill}$$
$$\text{Negative Charge} \longrightarrow \text{Potential Uphill}$$

Conclusion:

- Work is done by the system
- Potential Energy decreases for the system
- Kinetic Energy increases for the system

Incepting Ideas (1)

Hint: Conserve mechanical energy

Q: A proton with a speed of 2.0×10^5 m s^{-1} enters a region of space in which there is an electric potential. What is the proton's kinetic energy after it moves through a potential difference of 100 V? Will it gain or lose kinetic energy? Is the proton following a potential uphill or downhill?

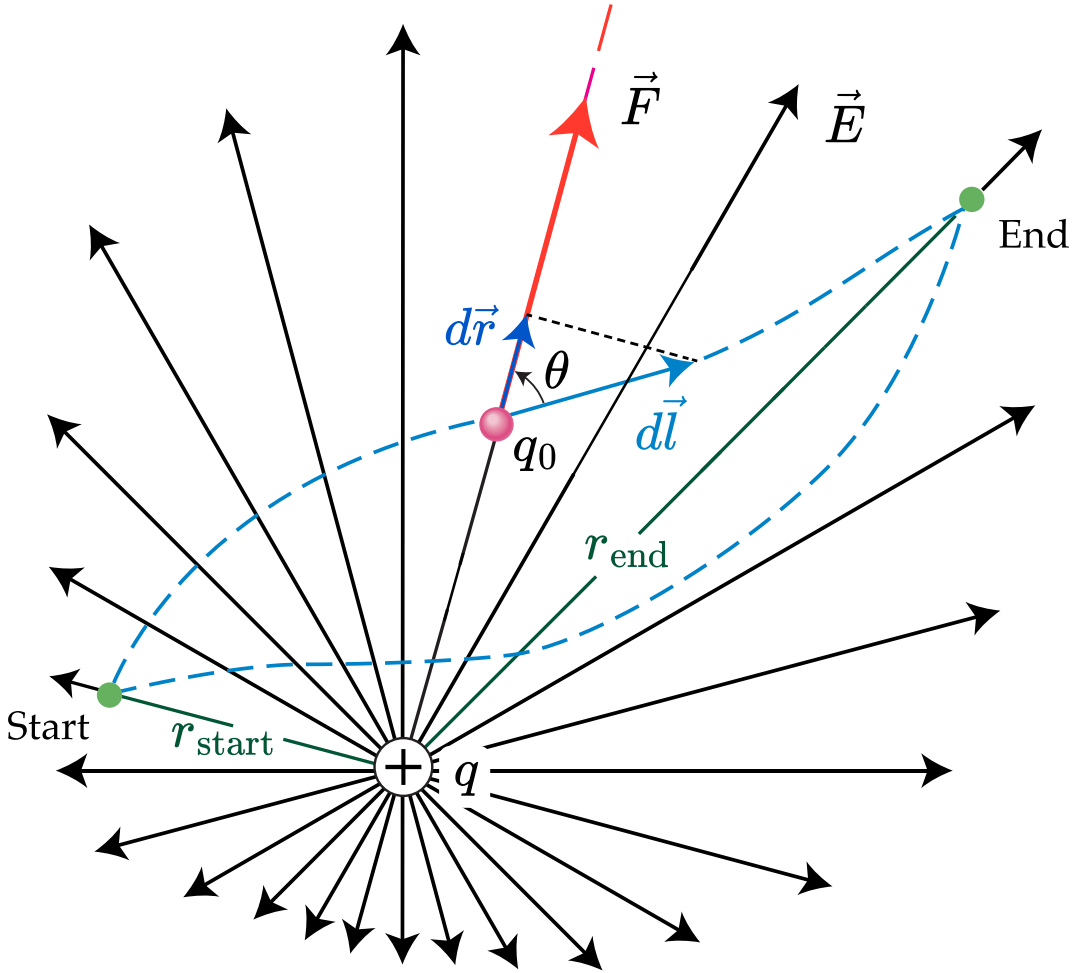
Alternative parameters to look at

Q: An electron with a speed of 2.0×10^5 m s^{-1} enters a region of space in which there is an electric potential. What is the electron's kinetic energy after it moves through a potential difference of 100 V? Will it gain or lose kinetic energy? Is the electron following a potential uphill or downhill?

Potentials of Non-Uniform Sources

Potential Change in a non-uniform Electric Field

Example: A point charge



Potential Change in a non-uniform Electric Field

Example: A point charge

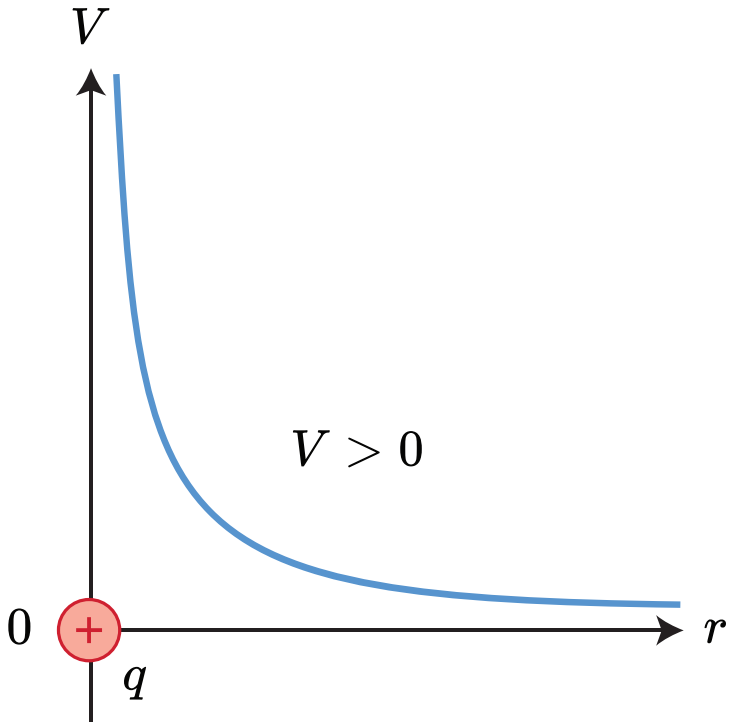
Physically, Electric Potential is the work done per unit charge ($+1$ C) to bring a charge from infinity (where the electric field is considered to be zero) **to a particular location r in an electric field**.

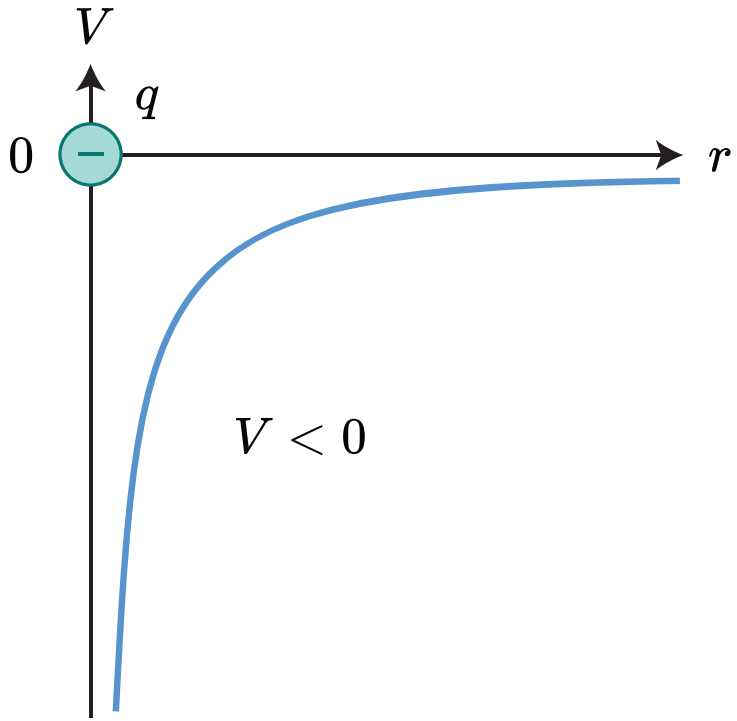
$$V_\infty = \frac{U_\infty}{q_0} = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

The Potential of Point Charges

Physics is superior than magic!

This shape is actually funnel-shaped in 3D.





Potential of a System of Charges

Example: Similar to work done to assemble/disassemble a distribution

The potential due to a point charge (i.e., the work W needed per unit charge to bring external point charges from infinite separation to a separation r) is

$$V_\infty = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

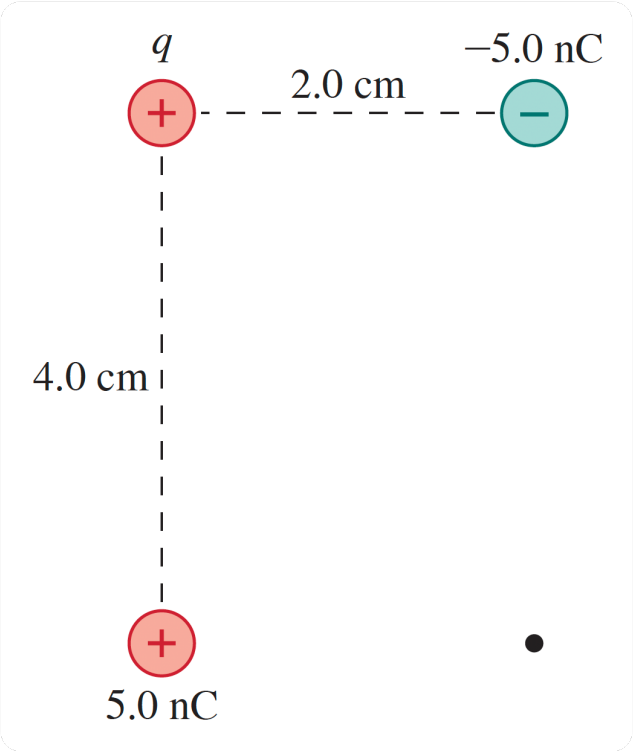
For a larger set of charges, the sum gives the potential energy

$$\begin{aligned} V &= V_1 + V_2 + V_3 + \ldots\ldots \\ &= \frac{1}{4\pi\epsilon_0}\sum_i \frac{q_i}{r_i} \end{aligned}$$

Incepting Ideas (4)

Hint: No vectors. No fuss.

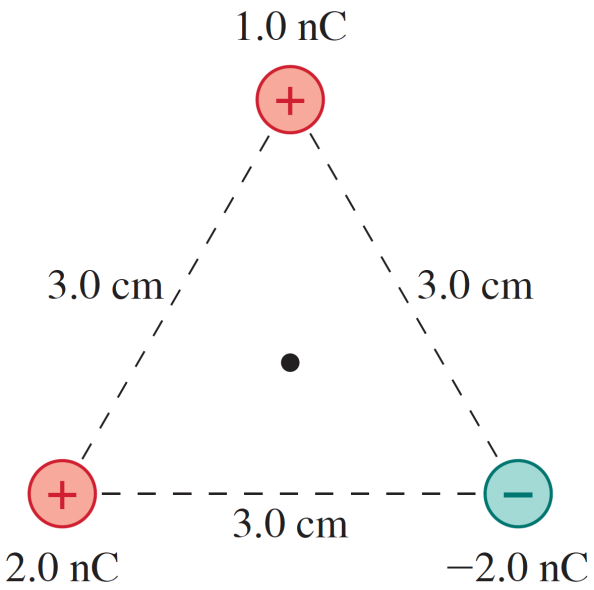
Q: The electric Potential at the dot is 3140 V. What is charge q?

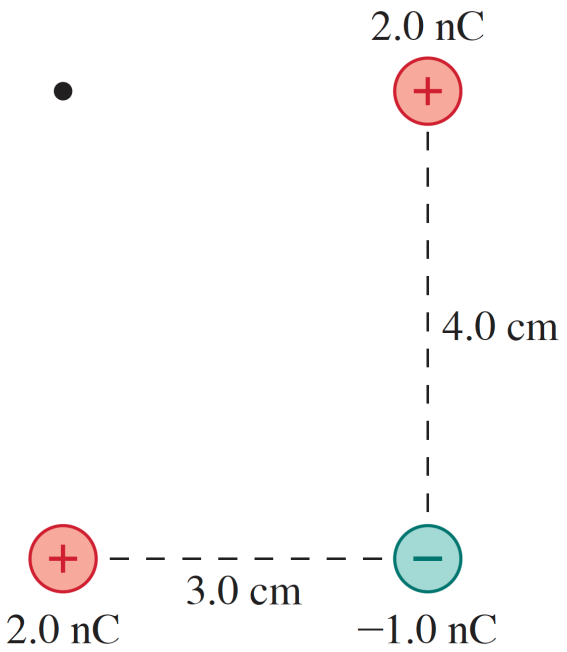


Incepting Ideas (5)

Hint: No vectors. No fuss.

Q: What is the electric potential at the point indicated with the dot?



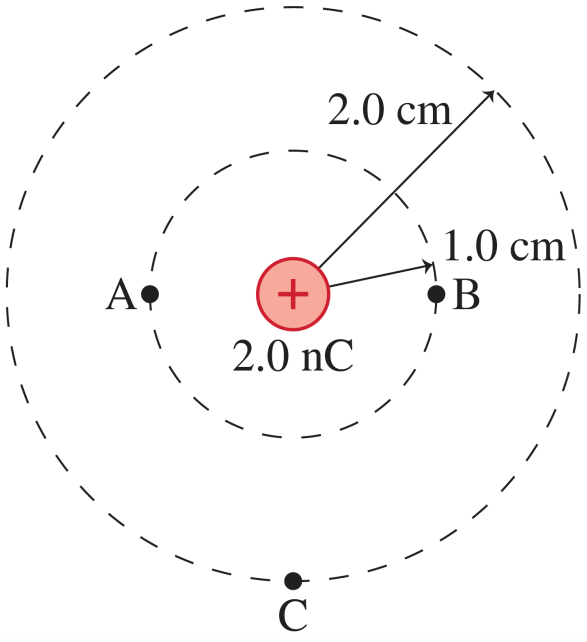


Hint: The incenter is $\frac{(\text{side length})}{\sqrt{3}}$ away from each charge at the vertices of the triangle.

Incepting Ideas (6)

Hint: Think 3D

Q: What is the electric potential at points (i) A, B, and C. (ii) What are the potential differences $\Delta V_{AB} = V_B - V_A$ and $\Delta V_{CB} = V_B - V_C$?



Incepting Ideas (7)

Hint: Practice! Practice! Practice!

Some Problems to Practice at Home on Electric Potential

Example 23.7, p-787 | Young-Freedman
Exercise Problem 23.12, p-801, 23.17, 23.20, p-802 | YF
Sample Problem 24.01, p-689, 24.03, p-696 | Resnick-Halliday
Exercise Problem 1, p-708, 6, p-709, 16, p-710, 17, 19, p-711 | RH

That is it for today!